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6 SEPTEMBER 1963
SUPERSEDING
MIL–C–12103B
20 JUNE 1960

MILITARY SPECIFICATION

COMPUTER, STADIA, CIRCULAR SCALE

This specification has been approved by the department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a plastic, circular slide rule graduated for stadia computations.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L–V–351 — Vinyl Chloride Polymer and Copolymer Rigid Sheets.

PPP–B–636 — Box, Fiberboard.

MILITARY

MIL–P–116 — Preservation, Methods of.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U. S. Military Property.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33d St., New York 16, N.Y.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1616 P Street, NW, Washington 6, D. C.)

3. REQUIREMENTS

3.1 Description. The stadia computer shall be of plastic with a square base (rounded corners optional) of 5⅞ inches plus or minus ¼ inch containing a base scale with graduations from 0.5 to 1000; and a rotatable, circular disc containing a difference in elevation scale and a horizontal distance scale graduated in degrees and functions thereof. The stadia computer shall be similar to the one shown in figure 1 and as specified herein.

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3.2 Material. Material shall be as specified herein.

3.2.1 *Plastic.* Plastic shall conform to Specification L–V–351, class 3.

3.2.2 *Fastener.* The fastener shall be a metal eyelet or other similar device.

3.3 Disc. The disc shall be concentrically attached to the base and shall rotate freely with a minimum amount of lateral motion. The disc shall be of a diameter that will overlap the graduations on the base $\frac{1}{16}$ inch.

3.4 Graduations. Graduation lines shall be clean-cut, clear, and of uniform width (at least 0.006 inch throughout the scale). Numerals and letters on the scale shall be clear, fadeproof, and of 4½- to 14-point type size. Graduations shall be positioned to insure that the values derived from the stadia computer for a particular rod intercept shall be within plus or minus one-half of one percent of the mathematical derivation for the same rod intercept. The scale on the base shall be graduated in logarithms to base 10 of numbers 0.5 to 1000. The difference of elevation scale on the disc shall be graduated in logarithms to base 10, of one-half the sine of twice the angles from 3 minutes to 45 degrees. The horizontal distance scale on the disc shall be graduated to the logarithms base 10, of the cosine squared of the angles from 0 to 45 degrees.

3.5 Indentification markings. The stadia computer shall be identified in accordance with Standard MIL–STD–130. The identification markings should be placed on the back side of the stadia computer except the words "STADIA COMPUTER" shall be printed across the top of the front side of the base in 12 or 14 point type.

3.6 Instructions. The following information shall be printed on the computer disc in a convenient location in 4½ to 14-point type size similar to figure 1:

DIRECTIONS FOR USE
with Example

Set disc zero (0) opposite rod reading (537) on outer scale.

Read Difference of Elevation (113½) on outer scale opposite Vertical Angle (12°30′) on disc.

Read Horizontal Distance (512) on outer scale opposite Vertical Angle (12°30′) on disc.

Distance figures are in unit of rod graduations.

When a constant is used in measuring stadia, the quantity to be added may be taken from the following table:

C + F	5°		10°		15°		20°		25°		30°	
	HOR.	VERT.	HOR.	VERT.	HOR.	VERT.	HOR.	VERT.	HOR.	VERT.	HOR.	VERT.
0.80 Foot	0.79	0.07	0.78	0.14	0.77	0.20	0.75	0.27	0.72	0.34	0.69	0.40
0.90 "	0.89	0.08	0.88	0.15	0.87	0.23	0.85	0.30	0.82	0.38	0.78	0.45
1.00 "	0.99	0.09	0.98	0.17	0.97	0.26	0.94	0.34	0.91	0.42	0.87	0.50
1.10 "	1.09	0.09	1.08	0.19	1.06	0.28	1.03	0.37	1.00	0.46	0.95	0.55
1.20 "	1.19	0.10	1.18	0.21	1.16	0.31	1.13	0.41	1.09	0.51	1.04	0.60

3.7 Workmanship. There shall be no splits in the plastic caused by riveting. The computer shall be free from warpage, cracks, nicks, burrs, rough edges, scratches, and other imperfections which affect the appearance or which might impair the serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure

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supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection of the computer.

(b) Inspection of preparation for delivery.

4.3 Quality conformance inspection.

4.3.1 *Sampling.* Sampling for examination and test shall be in accordance with Standard MIL–STD–105.

4.3.2. *Examination.* Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 1.5 percent defective.

4.3.3 *Test.* Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 1.5 percent defective.

4.4 Inspection procedure.

4.4.1 *Examination.* The computers shall be examined for the following defects:

101. Material not as specified.
102. Dimensions not as specified.
103. Identification markings not as specified.
104. Workmanship not as specified.
105. Graduations not as specified.
106. Instructions not as specified.

4.4.2 *Test.* The stadia computer shall be tested as follows. Set the zero (0) on the disc opposite the first assumed rod reading (position 1) in table I. The difference of elevation and the horizontal distance on the outer scale opposite the corresponding assumed vertical angle is then read and compared against the given values in table I. In turn, set the disc zero to the remaining positions and check against the distance and elevation tabular values. Evidence of any computed value from the stadia computer not within plus or minus 0.5 percent of the tabular value shall constitute failure of this test.

TABLE I. *Mathematically derived distance and elevation values*

	Assumed Rod Reading (R) (S = 100R)	Assumed Vertical Angle	Difference of Elevation S(½ sin 2A)	Horizontal Distance S cos²A
1	9.5	34°0′	4.4	6.5
2	110	5°6′	9.7	109
3	537	12°30′	113	512
4	850	11°20′	164	817

4.5 Inspection of preparation for delivery.

4.5.1 *Production inspection.*

4.5.1.1 *Unit of product.* For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 *Sampling.* Sampling for examination shall be in accordance with Standard MIL–STD–105.

4.5.1.3 *Examination.* Samples selected in accordance with 4.5.1.2 shall be examined as specified in 4.5.2.1. AQL shall be 2.5 percent defective.

4.5.2 *Inspection procedure.*

4.5.2.1 *Examination.* The preservation, packaging, packing, and marking shall be examined for the following defects:

101. Materials not as specified for levels A and B.
102. Marking illegible, incomplete, or incorrect for levels A, B, and C.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Each computer shall be preserved in accordance with Specification MIL–P–116, method 1C–1.

5.1.2 *Level C.* The computers shall be preserved and packaged in accordance with the supplier's standard practice.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* The computers shall be packed in quantities not to exceed 70 pounds in boxes conforming to Specification PPP–B–636, V3c, class 2, style optional. After closure, the boxes shall be waterproof sealed with tape and reinforced with bands of tape in accordance with the appendix to the box specification.

5.2.2 *Level B.* The computers shall be packed in close-fitting boxes conforming to Specification PPP–B–636, class 1, W5c, style optional. The gross weight of each box shall not exceed the weight limitation of the box specification. The boxes shall be closed in accordance with the appendix to the box specification.

5.2.3 *Level C.* The computers shall be packed to insure carrier acceptance and safe arrival at destination at lowest rates in containers complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

5.3 Marking. Packages and shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The computer is intended for use in computing horizontal distances and differences in elevation in stadia surveys.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Levels of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Stadia computer. Figure 1 shows a type of stadia computer which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other designs which conform to the specification.

Custodians:

Army—MO
Navy—Docks
Air Force—MAA

Preparing activity:

Army—MO (ERDL)

Project No. 6675–0193

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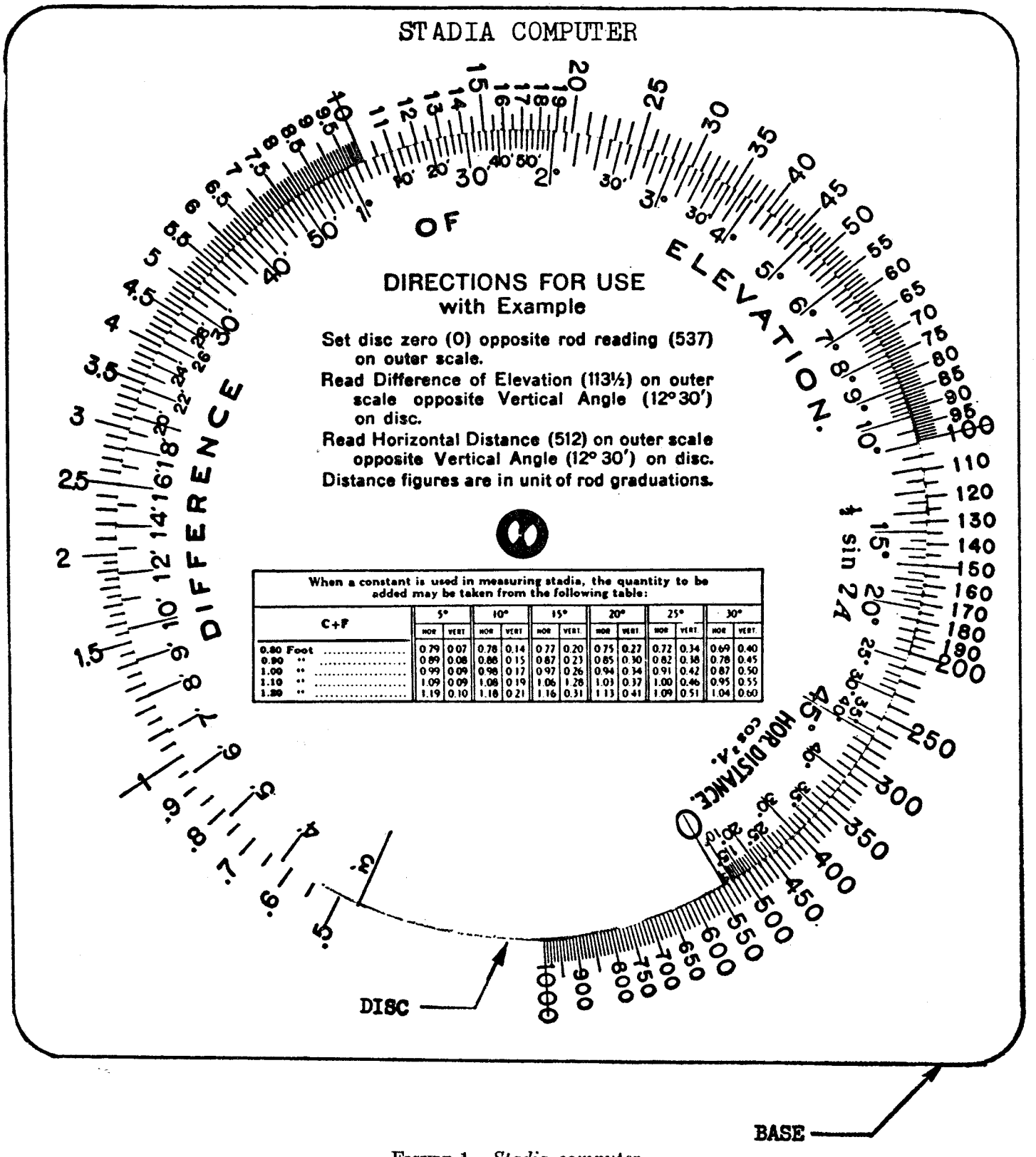


FIGURE 1. *Stadia computer.*

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